

Software Requirements Specification (SRS)

Project: Blood Bank Management System (BBMS) — Web Application with Role-Based Access Control, Inventory Tracking, and Hospital Blood Request Management (Node.js / Express / SQLite / Tailwind CSS)

Version: 1.0

Authors: Uttam (PES1UG24CS697), Abhinav K (PES1UG24CS701), Akshay Arcot (PES1UG24CS705), Navaneeth Tanuboddi (PES1UG24CS709) — Team T9

Date: 14-09-2026

Status: Draft for review

Revision history

Version	Date	Author	Change summary	Approval
0.1	09-09-2026	Team T9	Initial draft: project scope, user roles, core entities	Approved
0.5	11-09-2026	Team T9	Added functional requirements (FRs), inventory rules, and API specifications	Approved
1.0	14-09-2026	Team T9	Complete SRS with security requirements, UML use-case diagrams, test cases, and RTM	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Prof. Dept of CSE	coordinator.cse@pes.edu	14-09-2026
Course Instructor	Faculty Advisor	instructor.cse@pes.edu	14-09-2026
Team Lead	Navaneeth Tanuboddi	navaneeth.tanuboddi@gmail.com	14-09-2026

Table of Contents

- 1. [Introduction](#1-introduction)
- 2. [Overall description](#2-overall-description)
- 3. [External interface requirements](#3-external-interface-requirements)
- 4. [System features (detailed)](#4-system-features-detailed)
- 5. [Non-functional requirements (detailed)](#5-non-functional-requirements-detailed)
- 6. [Quality attributes & Acceptance tests](#6-quality-attributes--acceptance-tests)
- 7. [System models and diagrams (UML use-case)](#7-system-models-and-diagrams)
- 8. [Requirements Traceability Matrix (RTM)](#8-requirements-traceability-matrix-rtm)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Blood Bank Management System (BBMS), an enterprise-grade, role-based full-stack web application developed using Node.js, Express, SQLite, and Tailwind CSS. It defines the functional and non-functional requirements, external interfaces, security objectives and controls, system architectural models, and the verification criteria against which the delivered system will be assessed. The intended readers are the developer team (Team T9), reviewing faculty instructor, course coordinator, and laboratory evaluators.

1.2 Scope

The Blood Bank Management System encompasses the end-to-end operational lifecycle of voluntary blood donation, blood unit testing and processing, inventory management, hospital blood requisition, and emergency dispatch. The system provides role-based web portals for Donors, Hospital Representatives, Blood Bank Staff, and System Administrators.

Key in-scope subsystems include: donor pre-screening health questionnaires, appointment scheduling, digital donor cards, blood unit collection tracking with unique Bag IDs, mandatory serological infection screening (HIV, Hepatitis B/C, Syphilis, Malaria),

component separation (PRBC, Platelets, Fresh Frozen Plasma), First-Expired-First-Out (FEFO) stock monitoring, hospital requisition submission with clinical urgency prioritization, automated ABO/Rh cross-matching verification, cold-chain dispatch handover manifests, immutable audit logging, and a public availability dashboard.

Explicitly out of scope: Direct hardware automation of centrifuges/refrigeration IoT sensors, online monetary payment gateways (blood is donated voluntarily without commercial sale), and direct clinical transfusion administration inside operating theaters.

1.3 Audience

Developer Team (Team T9), Software Quality Assurance / Test Engineers, Course Instructors, Academic Evaluators, and Hospital Blood Transfusion Committees.

1.4 Definitions and Acronyms

Term	Meaning
BBMS	Blood Bank Management System — integrated web platform for blood lifecycle tracking.
RBAC	Role-Based Access Control — security architecture restricting access based on user role.
PRBC	Packed Red Blood Cells — red cell component used to treat acute hemorrhage and severe anemia.
FFP	Fresh Frozen Plasma — plasma frozen within 8 hours, containing labile coagulation factors.
Platelet Concentrate	Thrombocytes harvested to prevent bleeding in thrombocytopenic patients (5-day shelf life).
Serology Screening	Laboratory diagnostic assays for transfusion-transmissible infections (TTIs).
Quarantine	Holding status isolating blood units pending test results or discarding non-conforming units.
Cross-Matching	Compatibility testing between donor erythrocytes and recipient serum prior to issue.
FEFO	First-Expired-First-Out — inventory distribution strategy prioritizing closest expiration date.
RTM	Requirements Traceability Matrix — mapping requirements to modules, test cases, and status.

2. Overall description

2.1 Product perspective

The Blood Bank Management System operates as a centralized, self-contained client-server web application. The backend is powered by Node.js and Express RESTful services backed by an ACID-compliant SQLite relational database running in Write-Ahead Logging (WAL) mode for high concurrent read throughput. The frontend utilizes responsive semantic HTML5, vanilla modern JavaScript (ES6+), and Tailwind CSS for mobile-friendly UI rendering. The application is completely containerizable and can run locally or behind reverse proxies (Nginx, Cloudflare Tunnels) for secure HTTPS access.

2.2 Major product functions

- Donor registration, medical eligibility self-screening, and digital donor card generation.
- Blood donation appointment booking and blood drive camp scheduling.
- Blood collection logging with unique barcode-compatible Bag IDs and donor linkage.
- Laboratory serology testing entry with automated quarantine enforcement for reactive units.
- Blood component separation tracking (PRBC, Platelets, FFP) with distinct temperature and shelf lives.
- Real-time blood stock inventory management categorized by 8 blood groups and component types.
- First-Expired-First-Out (FEFO) inventory allocation and proactive expiration threshold alerts.
- Hospital blood requisition workflow supporting Emergency/STAT and routine priority queues.
- Automated ABO/Rh compatibility cross-matching engine.
- Cryptographically signed dispatch manifests and chain-of-custody delivery verification.
- Immutable system audit logging and regulatory compliance reporting.

2.3 User roles and characteristics

- **Public Donor** — Voluntary blood donor. Expects intuitive self-service portal, eligibility guidance, simple appointment booking, and instant access to digital donor cards.

- **Hospital Representative** — Authorized hospital physician or blood bank coordinator. Submits urgent blood requisitions, tracks fulfillment status, and confirms custody transfer.
- **Blood Bank Staff / Lab Technician** — Laboratory professional logging blood collections, recording infectious disease screening assays, separating components, and managing stock allocations.
- **System Administrator** — IT/Operations lead managing user roles, configuring blood bank parameters, inspecting audit trails, and generating regulatory compliance reports.
- **Course Evaluator / Inspector** — Academic evaluator reviewing source code modularity, test coverage, relational schemas, and adherence to Software Engineering standards.

2.4 Operating environment

- **Server:** Node.js 18+ runtime on macOS (Apple Silicon / Intel), Linux (Ubuntu 22.04 LTS), or Windows 10/11 Server.
- **Database:** SQLite 3.x embedded database with WAL mode enabled; zero external database daemon required.
- **Client:** Modern evergreen web browsers (Chrome 100+, Safari 15+, Firefox 100+, Edge 100+) on desktop, tablet, and mobile devices.
- **Network:** Standard HTTP/1.1 and HTTP/2 over TCP ports 3000 / 443 with TLS encryption.

2.5 Constraints and assumptions

- Standards compliance: Clean architectural separation between REST controllers, business service logic, and database access models.
- Relational integrity: Foreign keys strictly enforced across all database tables (donors, units, requests, logs).
- Safe dispensing: The system strictly blocks allocation of expired, un-tested, or serologically reactive blood units.
- Physical pre-condition: Clinical vitals (hemoglobin ≥ 12.5 g/dL, blood pressure, weight ≥ 45 kg) are verified by physical nursing staff at collection.
- Regulatory compliance: Complete audit logging of all inventory changes to fulfill National Blood Transfusion Council guidelines.

3. External interface requirements

3.1 User interfaces

The web interface is engineered with responsive, accessible Tailwind CSS components. Key screens include:

- **Public Landing & Availability Portal:** Real-time stock summary, donor educational guidelines, and blood camp schedules.
- **Donor Portal:** Clean health questionnaire, appointment selector with time-slot reservation, and printable digital donor card.
- **Staff Operations Dashboard:** Rapid unit logging with Bag ID scanner support, serology results checklist, and quarantine status toggles.
- **Hospital Order Portal:** Structured requisition form with blood group selectors, unit counts, priority indicators, and delivery tracking.
- **Admin Control Center:** Interactive stock analytics, user role administration, and immutable audit trail tables.

3.2 Hardware Interfaces

- **Client Workstation / Mobile Device:** Minimum 1024x768 resolution for staff dashboard; responsive down to 360px width for donor portal.
- **Barcode / QR Scanner:** Optional standard USB/Bluetooth HID keyboard emulation for rapid Bag ID and Donor Card scanning.
- **Host Server:** Standard computing hardware (minimum 1 vCPU, 1 GB RAM, 10 GB storage).

3.3 Software Interfaces

- **Operating System:** macOS, Linux (Debian/Ubuntu/CentOS), or Windows 10/11.
- **Runtime Environment:** Node.js (v18.x or v20.x LTS).
- **Database Engine:** SQLite3 embedded database engine.
- **Tunnel / Proxy:** Cloudflare Tunnel daemon / Nginx for reverse proxying and automated SSL/TLS termination.

3.4 Communications Interfaces

RESTful JSON APIs over HTTP/HTTPS. Stateful session management via HTTP-only, SameSite secure cookies and JWT tokens.

4. System features (detailed)

Each requirement below carries acceptance criteria and a reference test case. Functional requirement IDs follow the pattern BBMS-F.

4.1 User Authentication and Role Management

Description: Establish identity, enforce role-based access control (RBAC), and protect user sessions.

Req ID	Requirement (shall...)	Type	Prio.	Source / Stakeholder	Acceptance criteria / Test ref	Dependencies
BBMS-F-001	The system shall authenticate users via email and bcrypt-hashed passwords, generating secure session tokens.	Functional	High	Security / Admin	AC: Valid credentials grant access; invalid credentials return 401 Unauthorized. Test: TC-AUTH-01	User database
BBMS-F-002	The system shall enforce RBAC across Donor, Hospital, Staff, and Admin roles, returning HTTP 403 for unauthorized routes.	Functional	High	Security / Evaluator	AC: Donors cannot access staff endpoints; hospitals cannot alter inventory. Test: TC-AUTH-02	BBMS-F-001
BBMS-F-003	The system shall temporarily lock out accounts after 5 consecutive failed login attempts within 15 minutes.	Functional	Medium	Security	AC: 5th failure triggers 15-min lockout; event logged to audit table. Test: TC-AUTH-03	BBMS-F-001
BBMS-F-004	The system shall provide a password reset flow utilizing time-limited (15 min) cryptographic tokens.	Functional	Medium	Donor / Hospital	AC: Expired or reused tokens are rejected with error diagnostic. Test: TC-AUTH-04	BBMS-F-001
BBMS-F-005	The system shall terminate sessions and invalidate authentication tokens upon explicit user logout.	Functional	High	User / Security	AC: Subsequent requests with cleared session return 401 Unauthorized. Test: TC-AUTH-05	BBMS-F-001

4.2 Donor Management and Appointment Scheduling

Description: Facilitate donor registration, medical eligibility self-screening, and appointment management.

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
BBMS-F-006	The system shall capture donor profile details (name, DOB, blood group, contact, emergency contact).	Functional	High	AC: Profile stored with unique Donor ID; duplicate emails rejected. Test: TC-DNR-01	Donor portal
BBMS-F-007	The system shall present a mandatory clinical eligibility questionnaire (age, weight, hemoglobin, travel).	Functional	High	AC: Ineligible responses flag donor and block immediate appointment. Test: TC-DNR-02	BBMS-F-006
BBMS-F-008	The system shall enforce a minimum 90-day interval between whole blood donation appointments.	Functional	High	AC: Appointment booking disabled if prior donation < 90 days ago. Test: TC-DNR-03	Donation history
BBMS-F-009	The system shall generate a digital donor card displaying donor photo placeholder, blood group, and QR code.	Functional	Medium	AC: Digital card renders correctly and QR encodes valid donor verification URL. Test: TC-DNR-04	BBMS-F-006
BBMS-F-010	The system shall allow donors to schedule, view, and cancel upcoming appointments up to 24 hours prior.	Functional	Medium	AC: Slot reservations update camp capacity counts in real time. Test: TC-DNR-05	BBMS-F-008

4.3 Blood Collection, Testing, and Inventory Processing

Description: Record collected blood, enforce serology testing, quarantine infectious units, and separate components.

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case	Dependencies / Comments
--------	------------------------	------	----------	---------------------------------	-------------------------